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Case Study 11 : The Telemedicine Billing Engine

Source Code :

```
1. abstract class Plan {
2.     private String patientName;
3.     private double baseFee;
4.
5.     Plan(String patientName, double baseFee) {
6.         this.patientName = patientName;
7.         this.baseFee = baseFee;
8.     }
9.
10.     public String getPatientName() {
11.         return patientName;
12.     }
13.
14.     public double getBaseFee() {
15.         return baseFee;
16.     }
17.
18.     abstract double calculateBill();
19. }
20.
21. class SilverPlan extends Plan {
22.     SilverPlan(String patientName, double baseFee) {
23.         super(patientName, baseFee);
24.     }
25.
26.     @Override
27.     double calculateBill() {
28.         return getBaseFee() + 15;
29.     }
30. }
31.
32. class GoldPlan extends Plan {
33.     GoldPlan(String patientName, double baseFee) {
34.         super(patientName, baseFee);
35.     }
36.
37.     @Override
38.     double calculateBill() {
39.         return getBaseFee() + (getBaseFee() * 0.10) - 20;
40.     }
41. }
42.
43. public class TelemedicineBilling {
44.     static void printInvoice(Plan p) {
45.         System.out.println("Patient: " + p.getPatientName());
46.         System.out.printf("Final Bill: $%.2f%n", p.calculateBill());
47.         System.out.println();
48.     }
```

```
49.
50.     public static void main(String[] args) {
51.         Plan p1 = new SilverPlan("John", 100);
52.         Plan p2 = new GoldPlan("Sophia", 200);
53.         Plan p3 = new GoldPlan("Max", 50);
54.
55.         printInvoice(p1);
56.         printInvoice(p2);
57.         printInvoice(p3);
58.     }
59. }
60.
```

Program Output :

```
PS J:\COLLEGE_STUDY\Programming in JAVA\13_10_2026_CS__11_11> javac TelemedicineBilling.java
PS J:\COLLEGE_STUDY\Programming in JAVA\13_10_2026_CS__11_11> java TelemedicineBilling.java
Patient: John
Final Bill: $115.00

Patient: Sophia
Final Bill: $200.00

Patient: Max
Final Bill: $35.00
```